UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECD2523 DATABASE

Topic 9|NoSQL

Content adapted from Connolly, T., Begg, C., 2015. Database Systems: A Practical Approach to Design, Implementation, and Management, Global Edition. Pearson Education.

LECTURE LEARNING OUTCOME

By the end of this lecture, students should be able to:

- **Able to understand about NoSQL and particularly MongoDB with its core concepts.**
- **Able to understand about MongoDB ecosystem and available tools.**
- **Able to perform basic operations with MongoDB using MongoDB Shell and Compass.**
- **Able to implement the use of MongoDB in Python.**

01 Introduction to NoSQL and MongoDB

02 Core Concepts of MongoDB

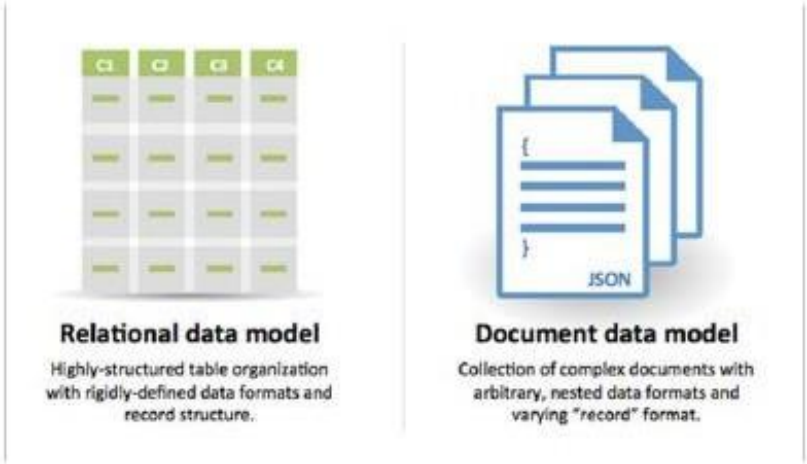
03 MongoDB environment and tools

NoSQL

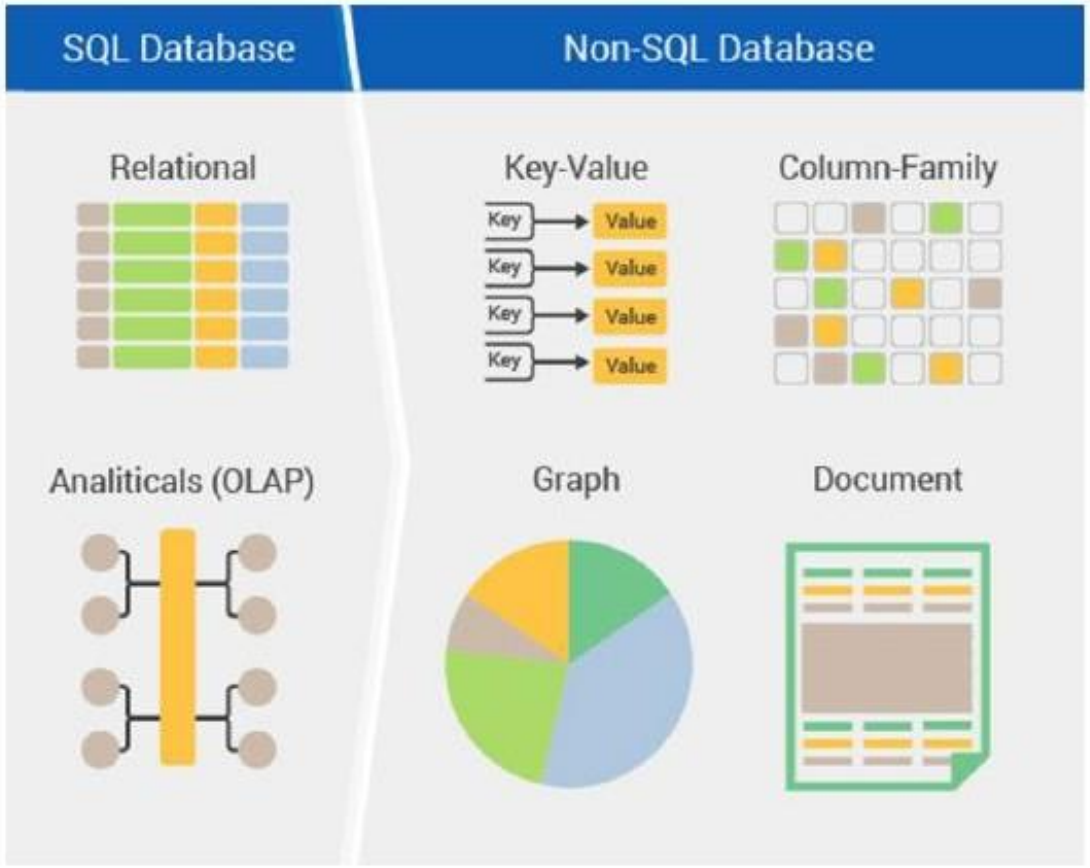
- NoSQL is a category of database systems that provides a **flexible** and **scalable** approach to storing and retrieving data, **differin**g from traditional relational database management systems (RDBMS). NoSQL databases are designed to **handle large volumes of unstructured, semi-structured, or structured data** and are often optimized for modern applications requiring high performance, horizontal scalability, and flexibility.

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SQL vs NoSQL

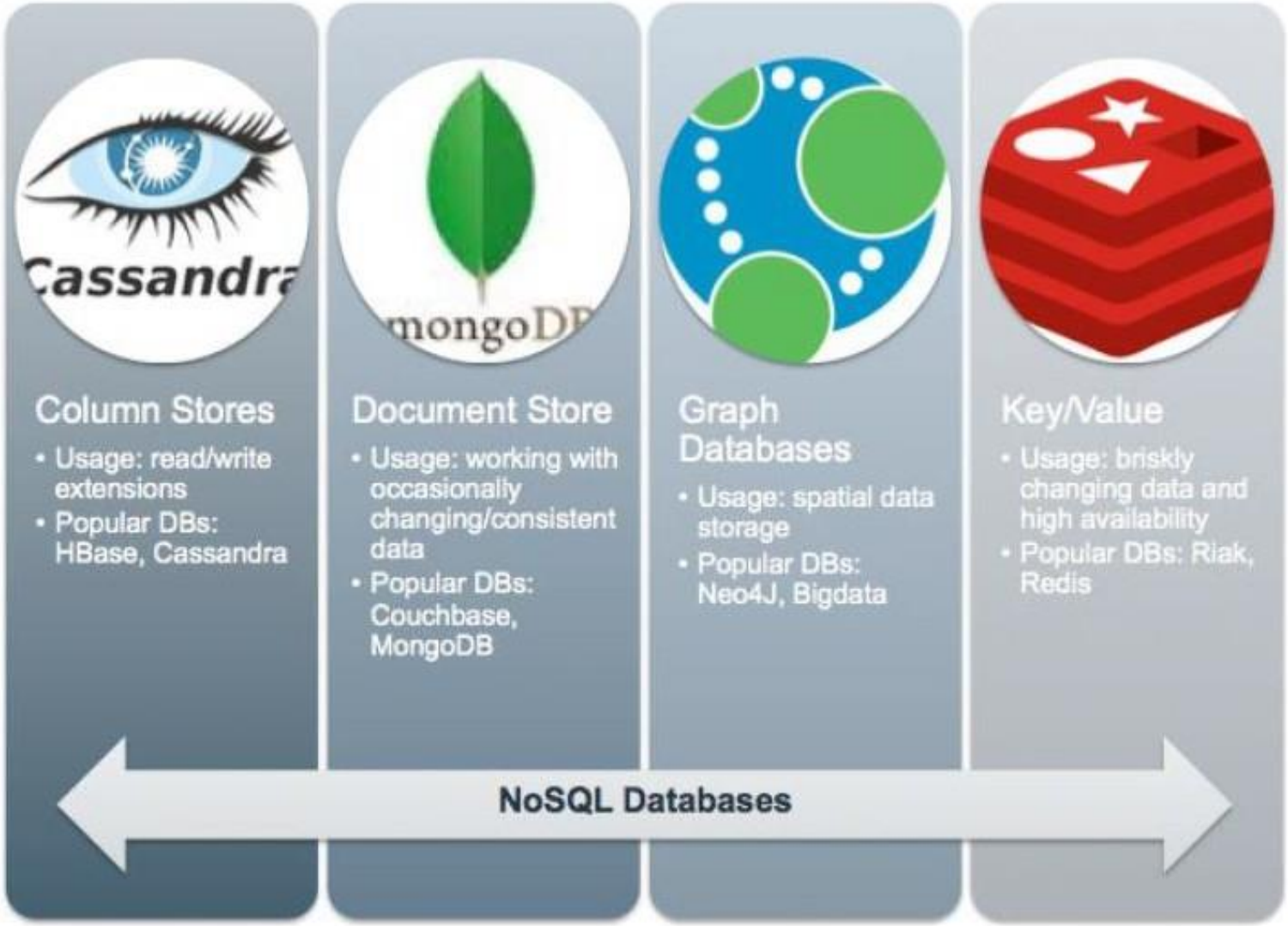






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SQL vs NoSQL



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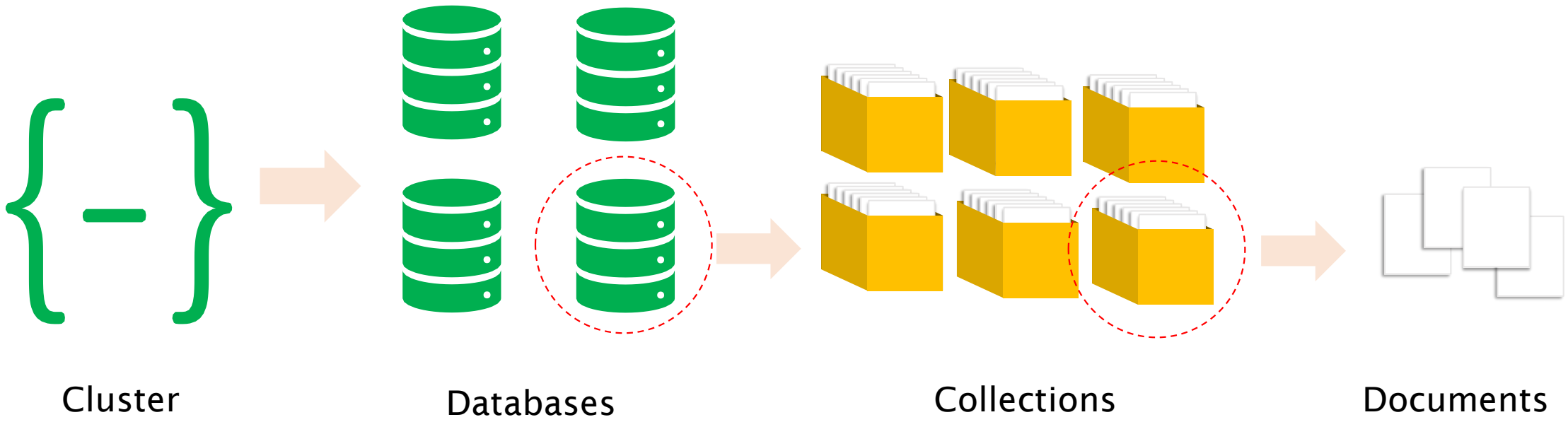
SQL vs NoSQL

Feature	NoSQL	SQL
Schema	Flexible schema	Fixed schema (tables and columns)
Scalability	Horizontally scalable	Vertically scalable
ACID Compliance	Often eventual consistency	Strong ACID compliance
Data Model	Varied (document, key-value)	Relational
Query Language	Database-specific APIs	SQL

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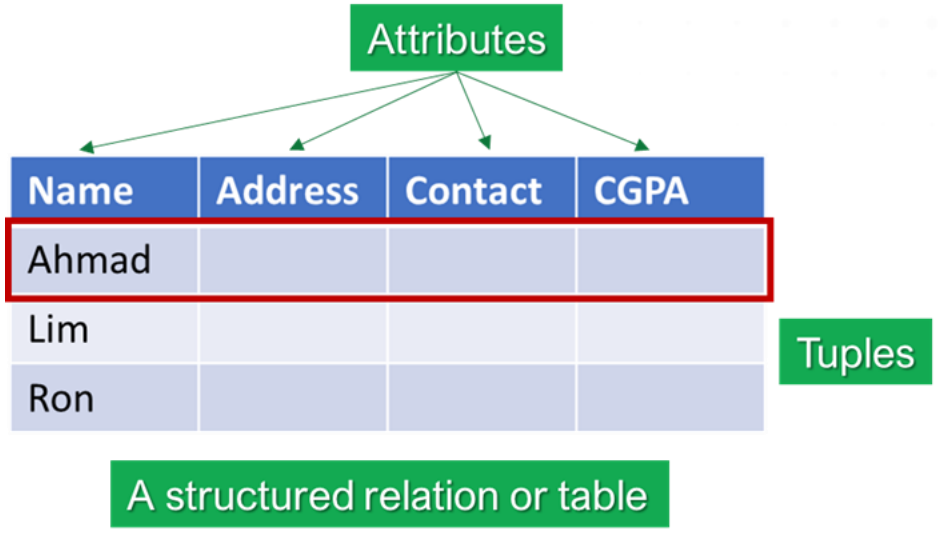
Getting familiar with MongoDB

- MongoDB is a NoSQL document-based database
- In a MongoDB cluster, there can be multiple databases, in each databases there can be multiple collections (imagine as Table in SQL) and each collections contains multiple documents (imagine rows of records).

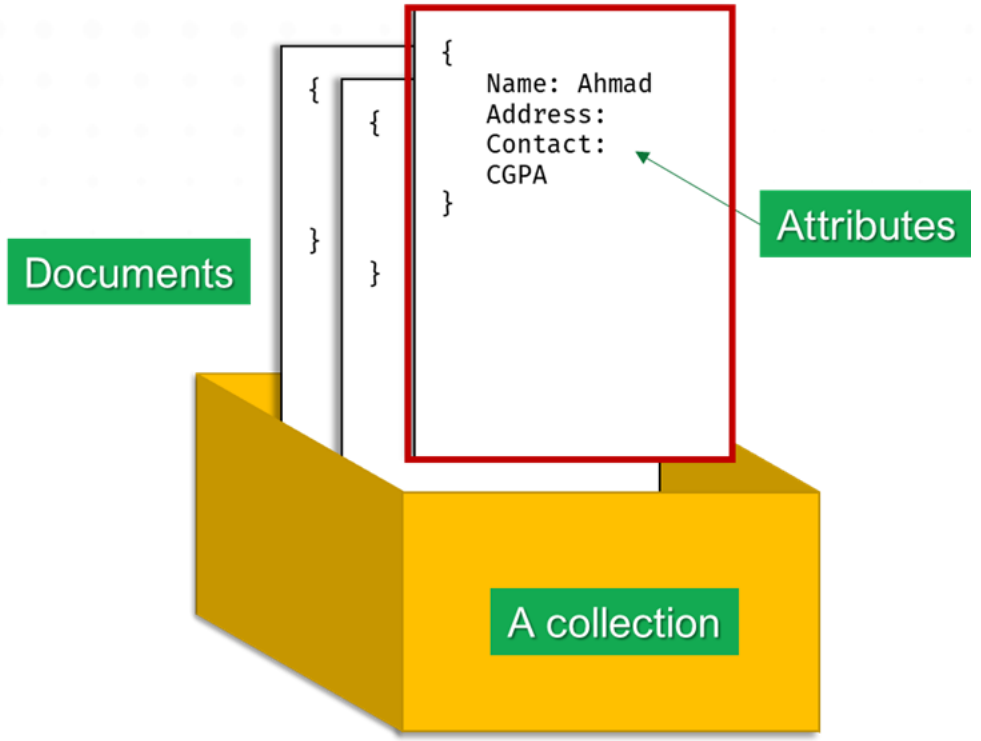


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How MongoDB different with SQL?



How SQL model the data



How MongoDB model the data

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Data Modelling in MongoDB

- Unlike SQL databases, where you must determine and declare a table's schema before inserting data, MongoDB's collections, by default, do not require their documents to have the same schema.
- MongoDB represents relationship through following concepts:
  - Reference document (Normalized model)
    - Store relationship between data by including references from one document to another.
    - Like “tagging” the documents with an attribute to “link”
    - In general, these consider as normalized data
  - Embedded document (Denormalized model)
    - Embedded all related data into a single document.
    - In general, these consider as denormalized data.

Refs: https://docs.mongodb.com/manual/core/data-model-design/

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Normalized Model (Reference Document)

CONTACT

Matric	phone	email
A18CSXXXX	0123456789	ahmad@x.com
A17KEXXXX	0112233456	desmond@x.com
A16KTXXXX	0198765432	lim@x.com

STUDENT

Foreign Keys

Name	Matric No.
Ahmad	A18CSXXXX
Desmond	A17KEXXXX
Lim	A16KTXXXX

3 tables
3 tuples per table

USERGROUP

matric	usergroup
A18CSXXXX	Dev
A17KEXXXX	User
A16KTXXXX	Dev

1 collections
3 documents per user

User document

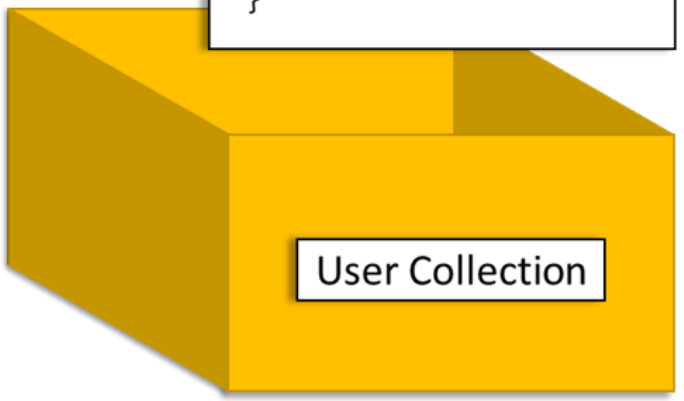
```
{
    _id: <ObjectID1>
    Matric: A18CSXXXX
    Name: Ahmad
}
```

Usergroup document

```
{
    _id: <ObjectID2>
    user_id: <ObjectID1>
    role: "Dev"
}
```

Contact document

```
{
    _id: <ObjectID3>
    user_id: <ObjectID1>
    phone: 0123456789
    email: "ahmad@x.com"
}
```



However, using reference document in MongoDB, in client-side application must issue follow-up queries to resolve the references

Simply said, more complex.

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Denormalized Model (Embedded document)

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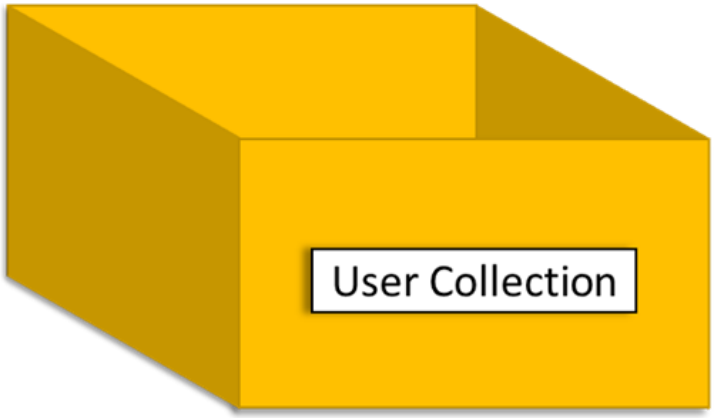
USERGROUP

matric	usergroup
A18CSXXXX	Dev
A17KEXXXX	User
A16KTXXXX	Dev

3 tables
3 tuples per table

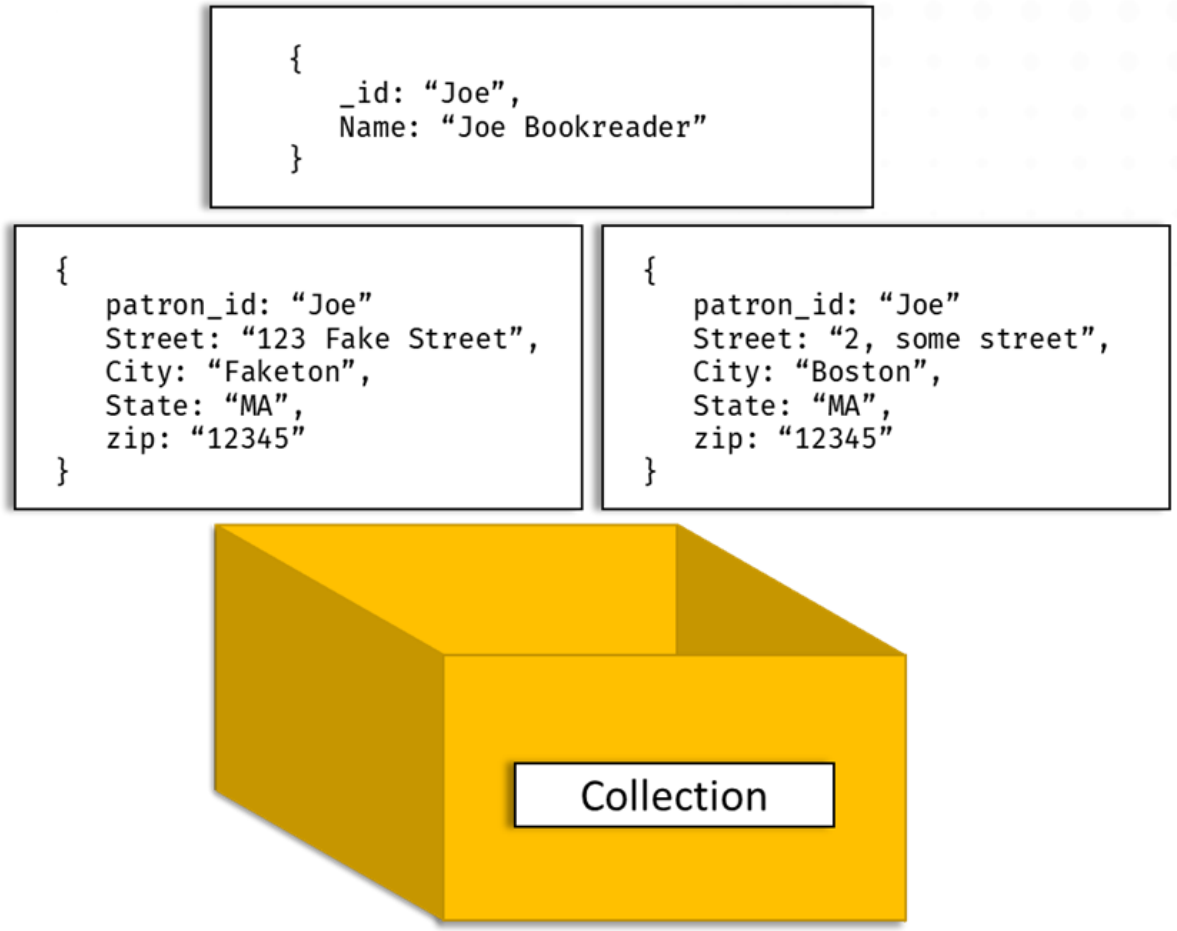
```
{
    _id: <ObjectID1>
    Matric: A18CSXXXX
    Name: Ahmad
    usergroup: {
                  role: "Dev"
               }
    contact: {
                  phone: 0123456789
                  email: "ahmad@x.com"
             }
}
```

1 collections
1 documents per user

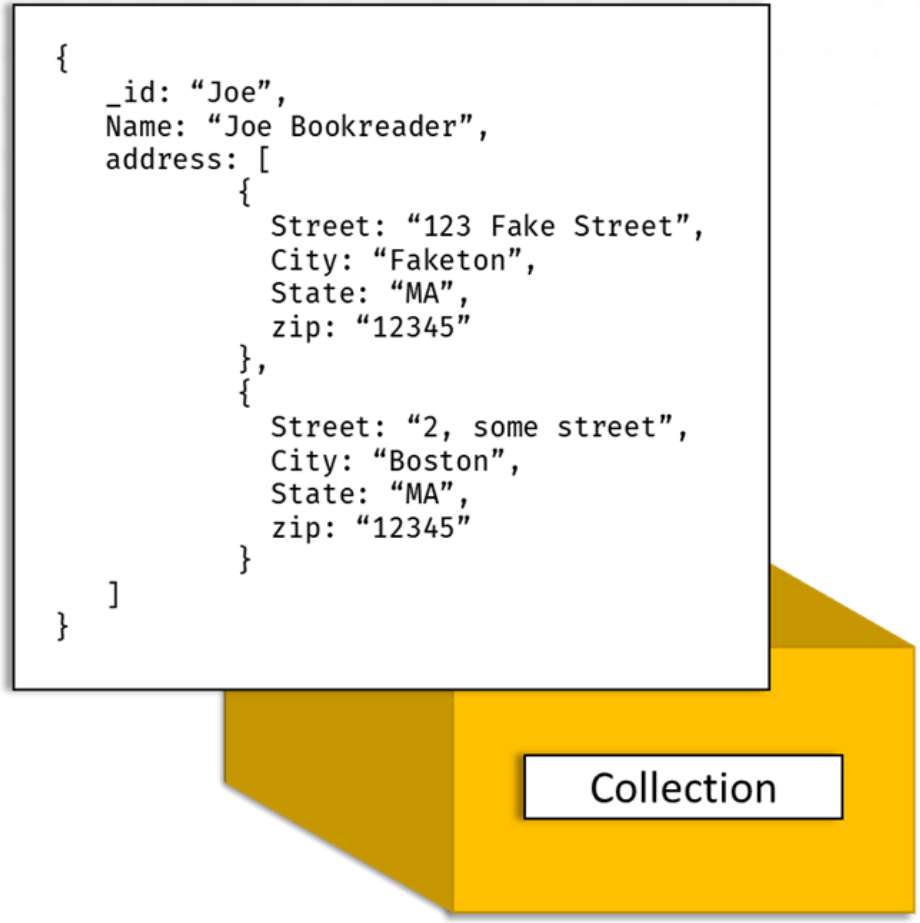


- Better performance in read operations,
- Simpler process in retrieving information from database,
- Possible to update related data in a single atomic write operation

Modelling One-to-Many Relationship



REFERENCE DOCUMENTS



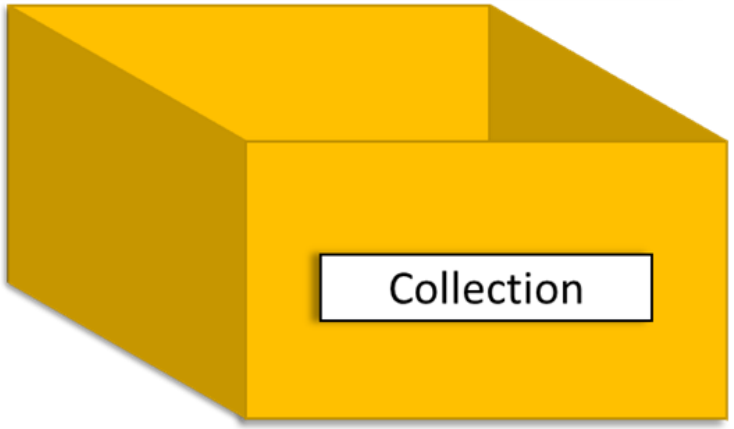
EMBEDDED DOCUMENTS

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Modelling One-to-One Relationship

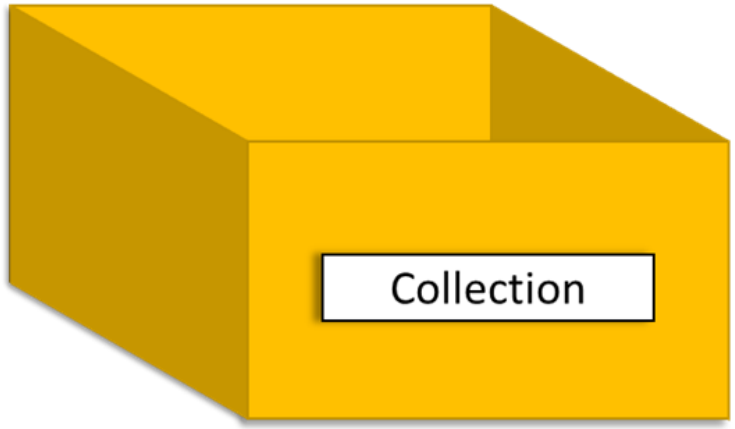
```
{
    _id: “Joe”,
    Name: “Joe Bookreader”
}
```

```
{
    patron_id: “Joe”
    Street: “123 Fake Street”,
    City: “Faketon”,
    State: “MA”,
    zip: “12345”
}
```



REFERENCE DOCUMENTS

```
{
    _id: “Joe”,
    Name: “Joe Bookreader”,
    address: {
                Street: “123 Fake Street”,
                City: “Faketon”,
                State: “MA”,
                zip: “12345”
              }
}
```



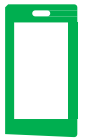
EMBEDDED DOCUMENTS

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MongoDB Environment and Tools



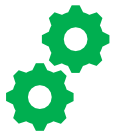
MongoDB Charts



MongoDB Realm

Cloud Servers
(MongoDB Atlas)





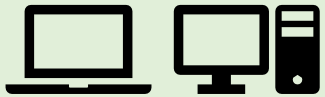
MongoDB Compass



MongoDB Shell



MongoDB BI
connectors



Local Servers
(MongoDB Community Server)

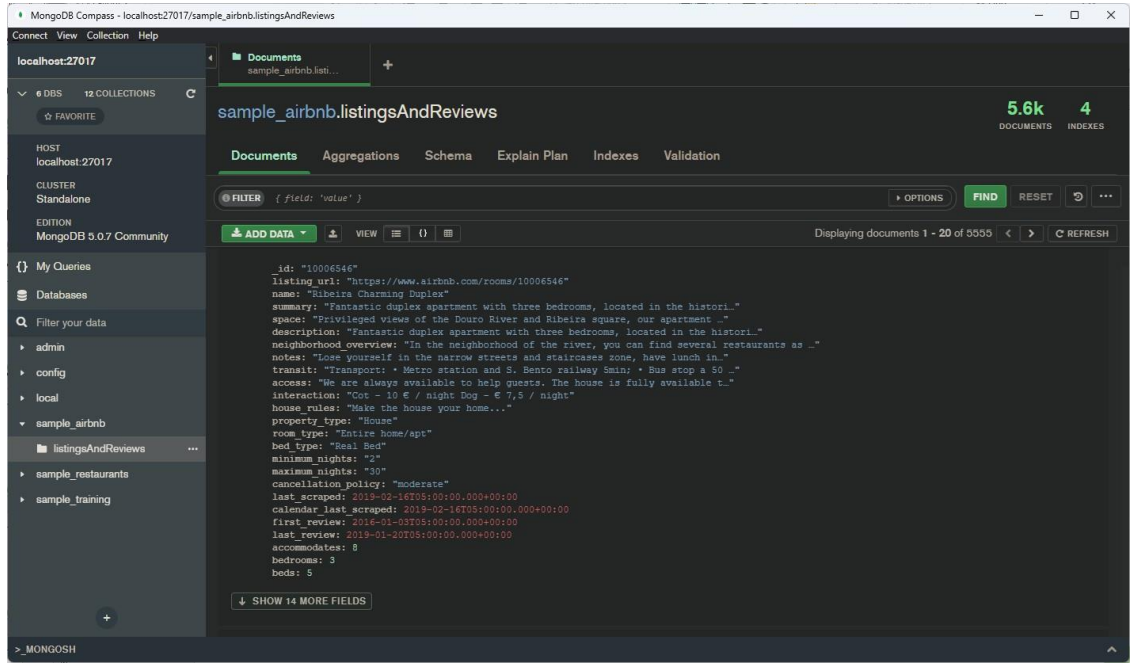


MongoDB
Database Tool

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Example: Using MongoDB Compass

- MongoDB Compass provide user friendly GUI to
  - Manage data in MongoDB
  - Do query, basic CRUD, aggregation
  - Provide descriptive analytics on the schema of the collection (not available in community edition)
  - Backup / restore
- MongoDB Compass can be used to connect and remotely manage MongoDB Atlas

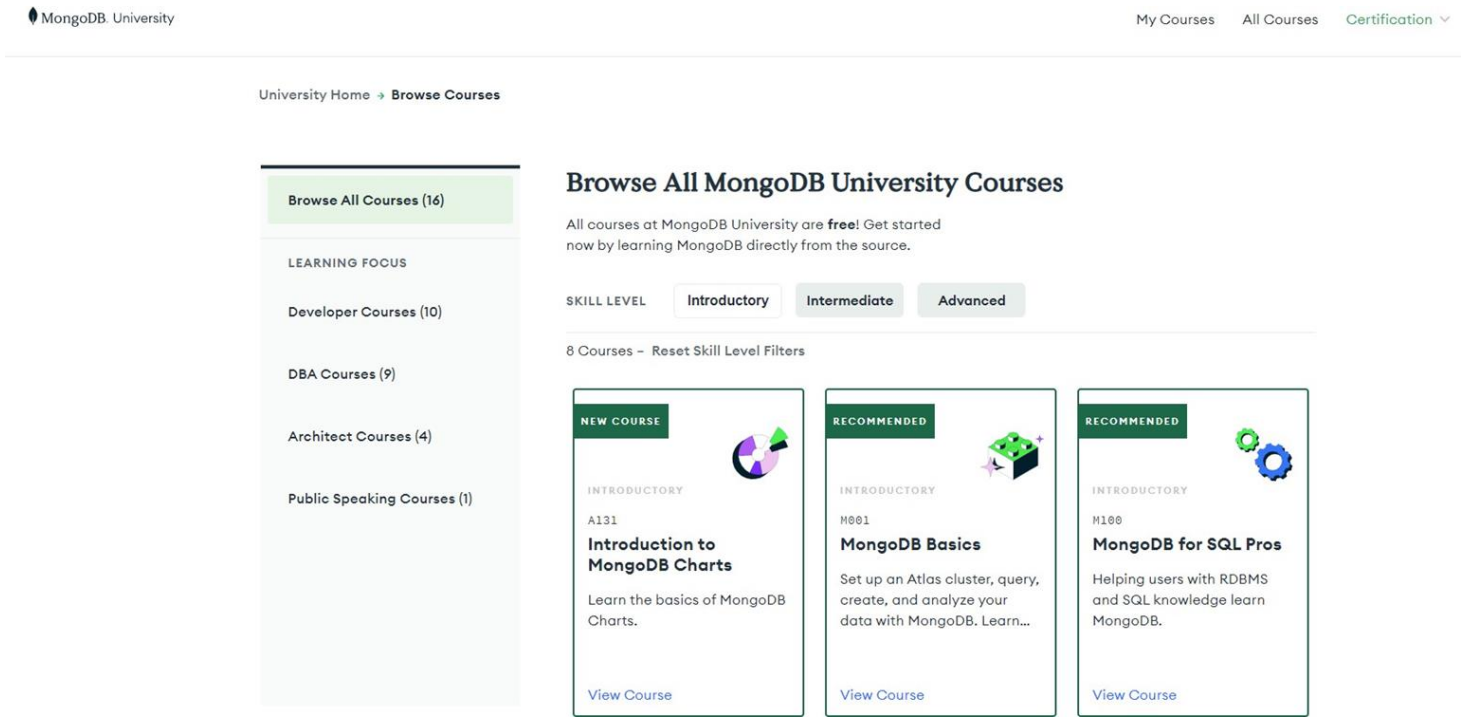


Resources

Where you can get more info and learn more!

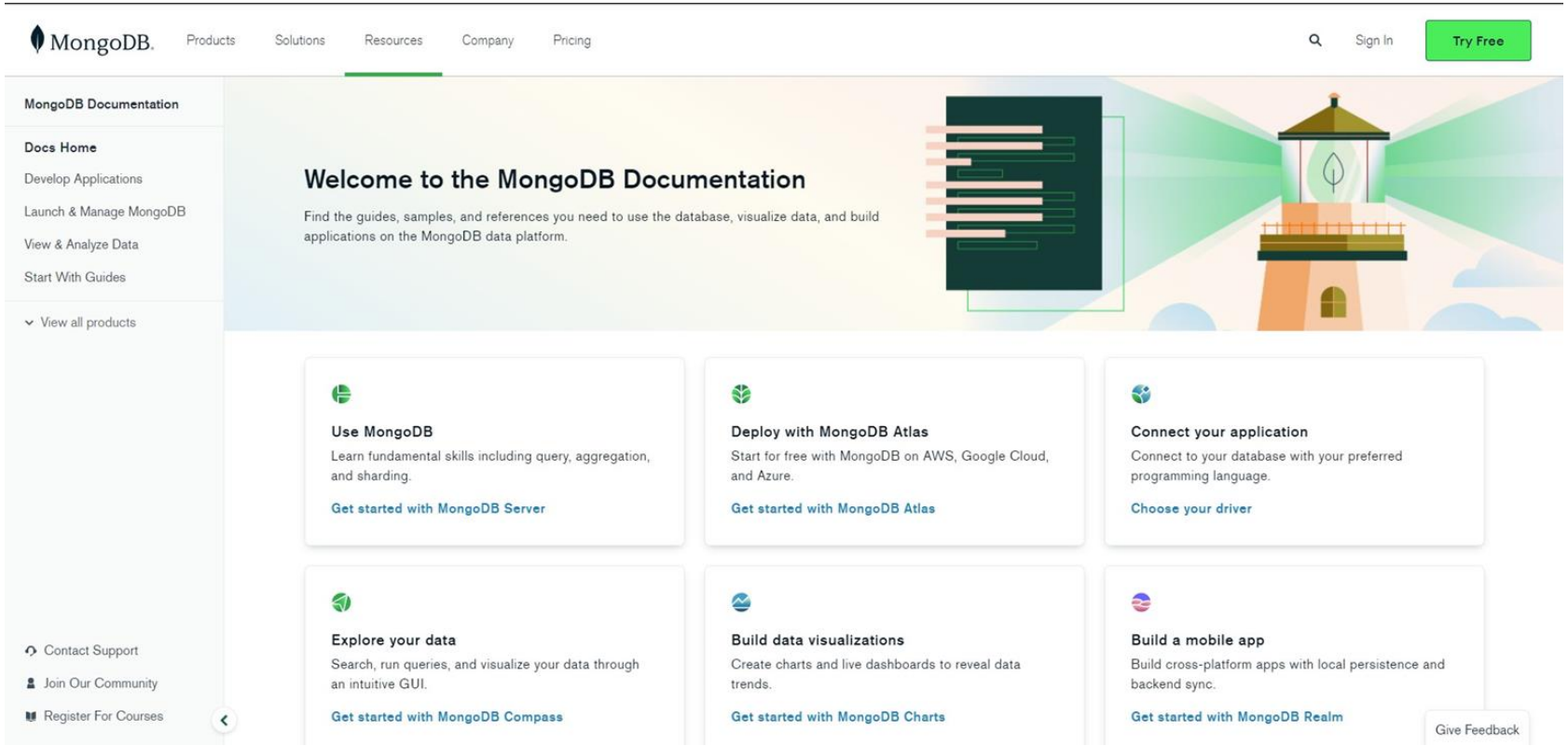
Resources & Learning Path

- MongoDB University offers many free courses regarding MongoDB ranging from introductory to advanced.
- https://university.mongodb.com/
- Professional certification also available if needed (not free)



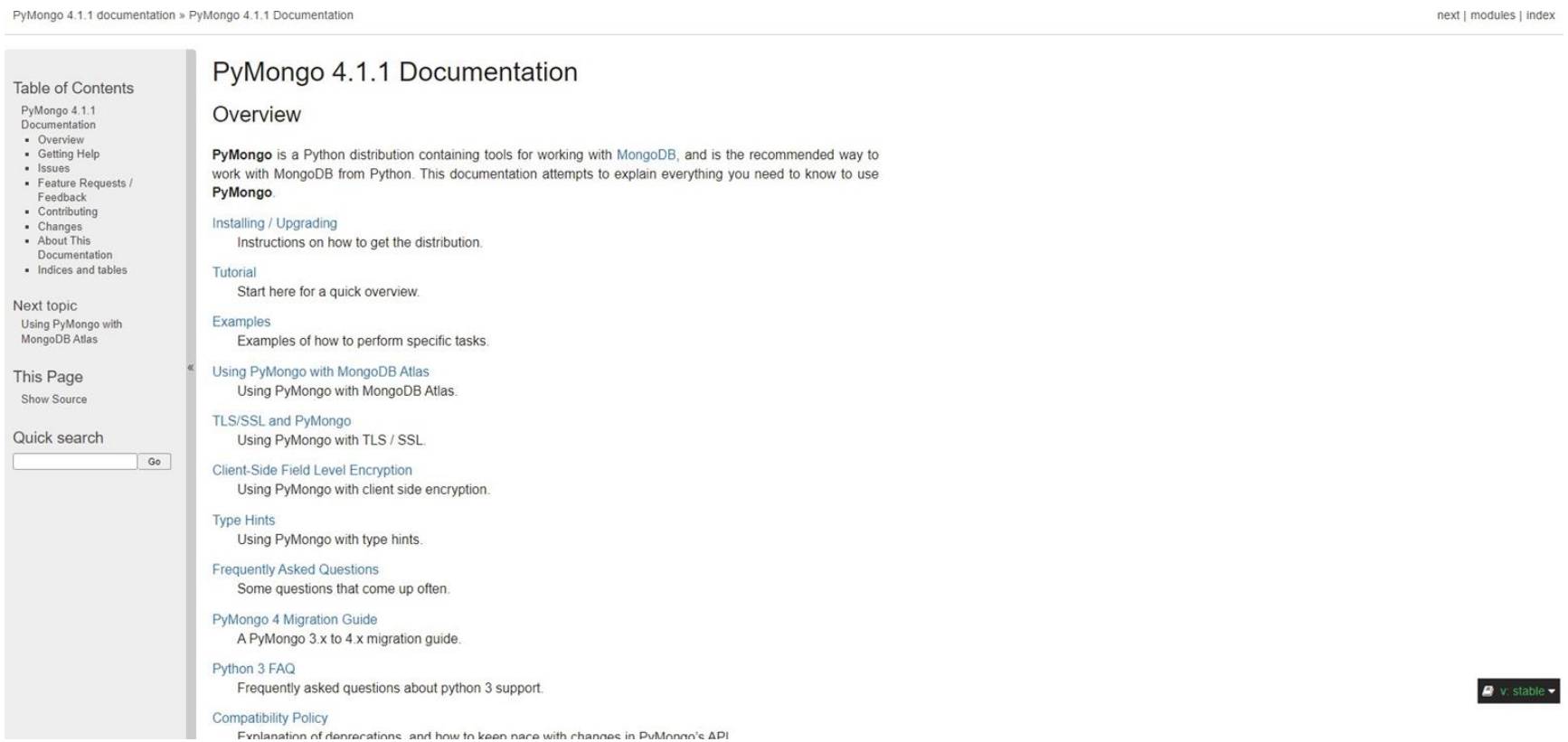
Resources & Learning Path

- MongoDB Documentation also provide many resources in using and managing MongoDB
- https://www.mongodb.com/docs/



Resources & Learning Path

- PyMongo documentation offers tutorials, examples to assist you in how to use MongoDB in Python.
- https://pymongo.readthedocs.io/en/stable/





Innovating Solutions
Menginovasi Penyelesaian